

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal DLA

Special Circulations (notes to go with online video)

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Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0144	Describe how the theory of metabolic regulation of blood flow accounts for active hyperemia and reactive hyperemia.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0145	Describe the phasic flow of blood to the ventricular myocardium through an entire cardiac cycle.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0146	Explain how arterio-venous O ₂ difference and oxygen extraction in the heart is unique when compared with other body organs.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0147	Explain the mechanism whereby the coronary blood flow is coupled to the myocardial workload and oxygen demand, and what type of stimuli can cause an increase in that flow
SOM.MK.II.BPM1.3.CPR.2.PHYS.0148	Explain the effects of the sympathetic nervous system stimulation on the heart rate, contractility, coronary vascular resistance, and coronary blood flow; and particularly during exercise, how the sympathetic stimulation has a direct and indirect effect determining the coronary blood flow
SOM.MK.II.BPM1.3.CPR.2.PHYS.0149	Explain the concept coronary vascular reserve, the role of collateral blood vessels, and the physiologic and pathological events that decrease the coronary vascular reserve
SOM.MK.II.BPM1.3.CPR.2.PHYS.0150	Discuss the mechanisms of regulation of the cerebral blood flow (myogenic, neurogenic, metabolic, and endothelial) and the importance of O ₂ , CO ₂ and pH in this process
SOM.MK.II.BPM1.3.CPR.2.PHYS.0151	Discuss the importance of intracranial pressure in the cerebral circulation
SOM.MK.II.BPM1.3.CPR.2.PHYS.0152	Contrast local and neural control of cutaneous blood flow.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0153	Discuss the unique characteristics of skin blood flow that are adaptive for body temperature regulation.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0154	Contrast the systemic and pulmonary circulations with respect to pressures, resistance to blood flow, and response to hypoxia.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0155	Describe the cardiovascular consequences of exercise on peripheral resistance, and cardiac output
SOM.MK.II.BPM1.3.CPR.2.PHYS.0156	Contrast the neural and local control of skeletal muscle blood flow at rest and during exercise.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0157	Discuss the interaction of a) intrinsic (local), b) neural, and c) humoral control mechanisms and contrast their relative dominance in the CNS, coronary, cutaneous, and skeletal muscle vascular beds.

Assigned Readings

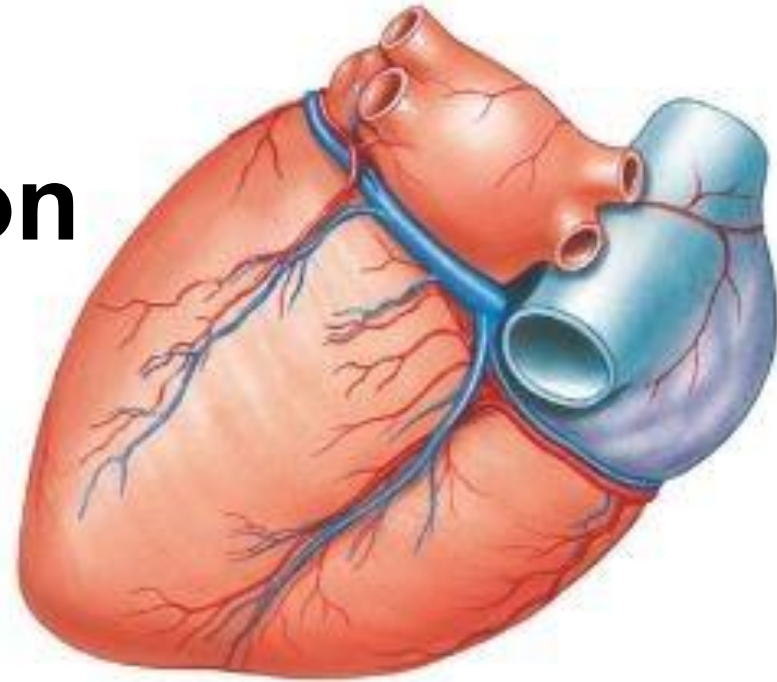
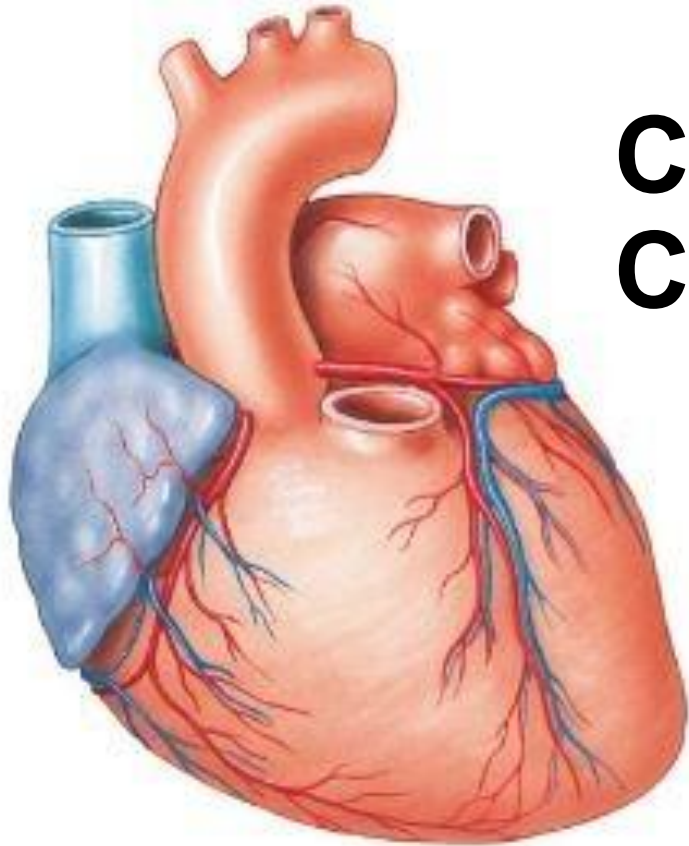
- Rhoads & Bell Medical Physiology. 6th ed. Lippincott. Chapter 16: Special Vascular Physiology of Individual Organs.
- Cardiovascular Physiology Concepts, 3rd ed. Richard E. Klabunde. Chapter 7: Organ Blood Flow-Special Circulations.

Special Circulations discussed in this DLA

- Coronary circulation: Heart
- Cerebral circulation: Brain
- Cutaneous circulation: Skin
- Skeletal circulation: discussed in DLA on Exercise Physiology
- Pulmonary circulation: discussed as a separate DLA

Please revise DLA on Local Control of Blood Flow before reading this DLA.

Coronary Circulation

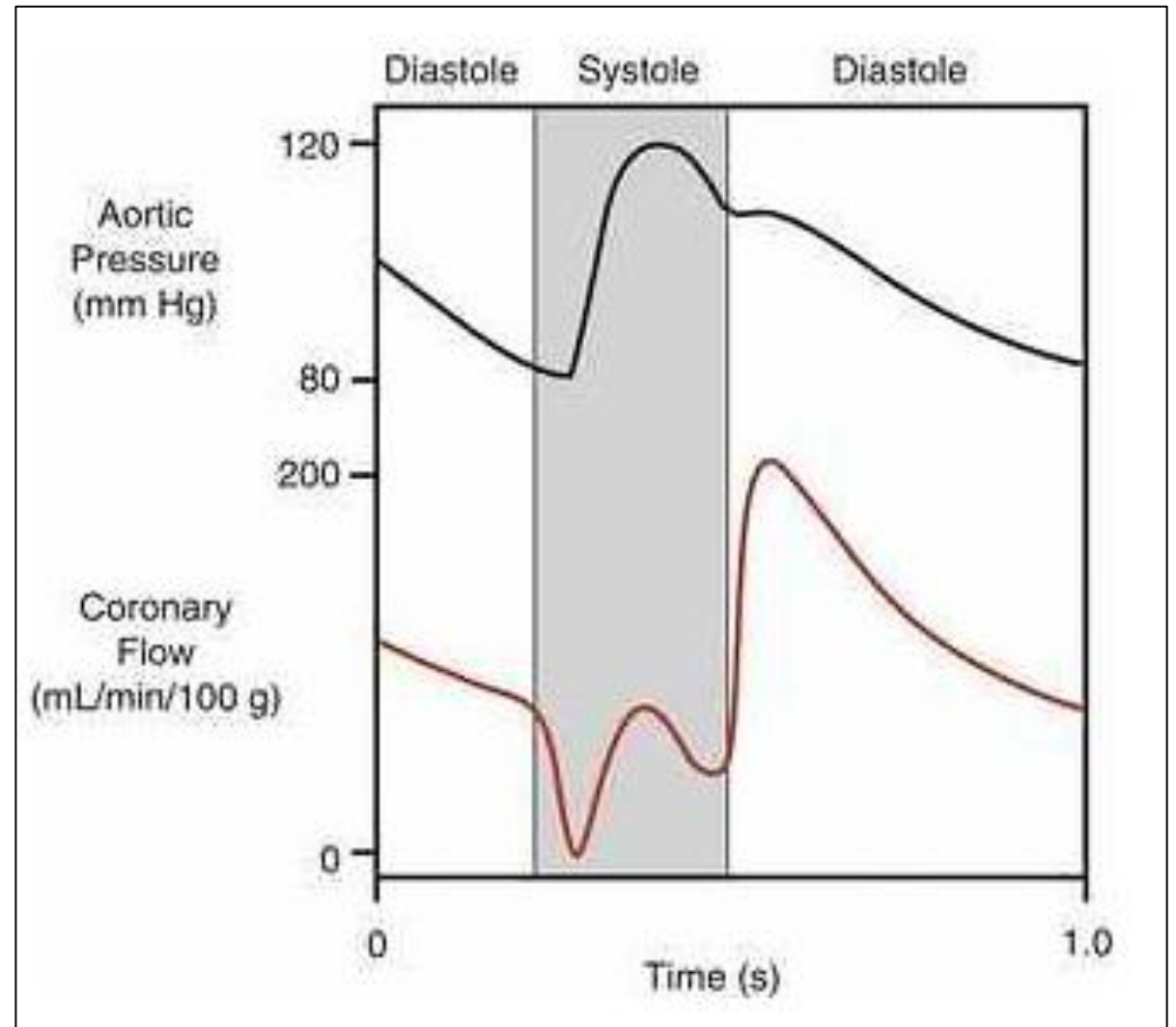


Coronary Circulation: Major Features

- The coronary circulation receives about 5% of cardiac output at rest.
- Oxygen extraction from blood is high at ~ 70% at rest.
- To get more O_2 to the heart, the blood flow must increase, hence O_2 delivery to heart is **flow-limited**.
- Major mechanism of coronary blood flow is by release of the local metabolites like **adenosine and nitric oxide** (both are vasodilators).
- Role of sympathetic nerves is less important as the effects of local metabolites override sympathetic vasoconstriction.
- Blockage of a major artery leads to ischemia downstream of the block because of **lack of collateral vascular connections**.

Coronary blood flow is pulsatile during cardiac cycle

- Blood flow is lower during systole because of **mechanical compression** of intramuscular coronary vessels.
- Flow is **maximal** during the early part of diastole, and then it falls as aortic pressure declines.

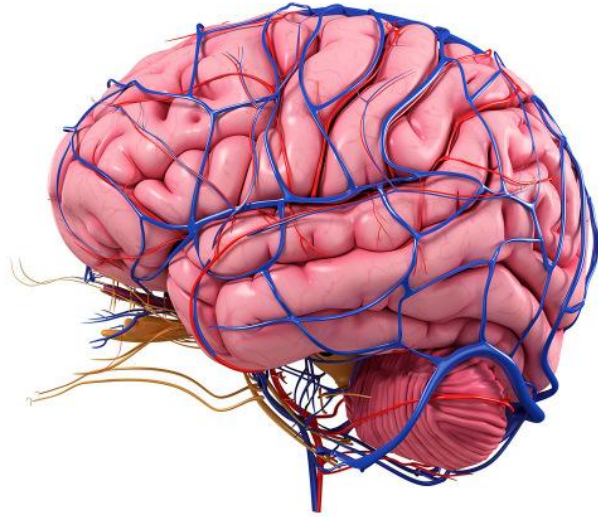


Relationship between Coronary Blood Flow, O₂ supply and O₂ demand

- **O₂ supply**: Amount of oxygen in the arterial blood reaching the myocardium. This depends on the amount of **blood flow**.
- **O₂ demand**: Amount of oxygen consumption by the myocardial tissue. This depends on the **metabolism**.
- Normally, the O₂ demand is matched with the supply by an adequate blood flow.
- In diseased states, either supply decreases or demand increases, or both. This leads to myocardial hypoxia and poor myocardial performance.
- Coronary circulation shows a **strong autoregulation** of blood flow and can maintain near-normal flows over a wide range of perfusion pressures. Both myogenic and local metabolites contribute.

Coronary Artery Disease: Atherosclerosis is the major cause

- Atheromatous plaques in coronary vessels reduce coronary blood flow.
- Decreased blood supply and tissue hypoxia under stress causes chest pain called **angina pectoris**. The pain is mostly because of release of adenosine.
- If the blood flow is completely blocked, the regional myocardial cells die resulting in **myocardial infarction** (MI).
- Both angina and MI produce progressive weakening of the heart and result in cardiac failure.
- Massive MI may lead to sudden cardiac death by cardiac arrhythmias and pump failure.

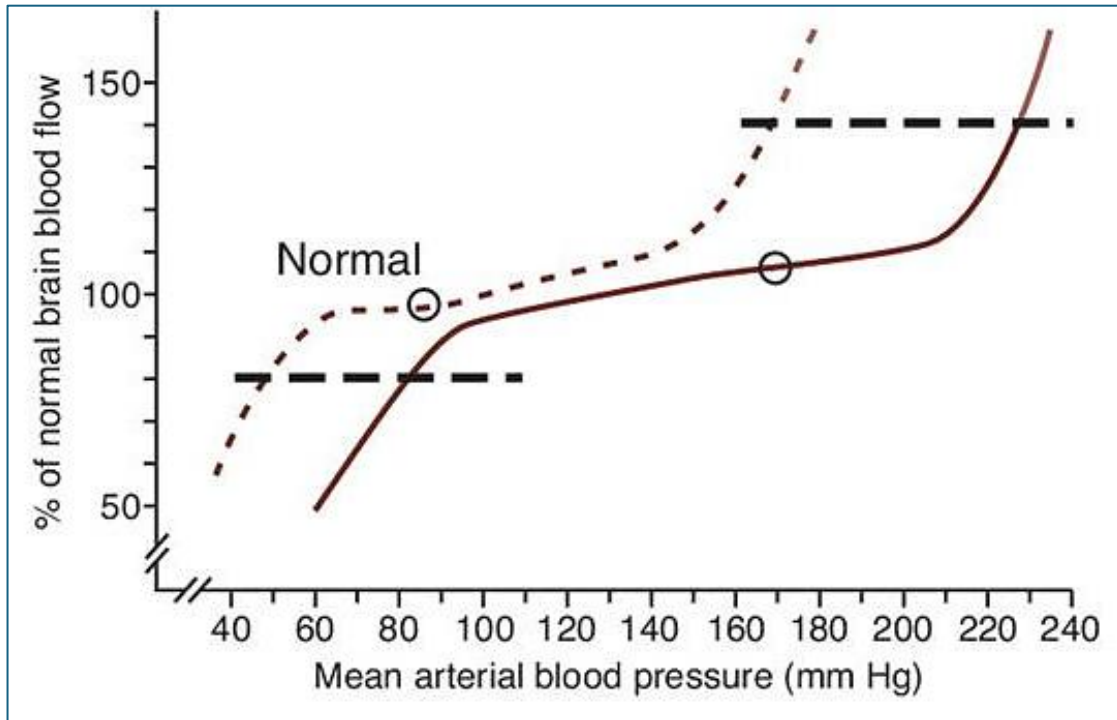


Cerebral Circulation

Cerebral Circulation: Major Features

- Cerebral circulation receives about 15% of cardiac output at rest.
- Critical to maintain a normal blood flow **all the time** as brain uses glucose as the major energy source.
- Disruption of flow for a few seconds leads to altered consciousness, but for a few minutes results in irreversible brain damage.
- Interconnecting arterial vessels at the base of the skull called, “**Circle of Willis**” helps cerebral perfusion if one of the vessels is occluded.
- Intracranial pressure plays a major role in cerebral blood flow.
- Cerebral vessels are highly sensitive to vasodilation by **CO₂ and H⁺**.

Autoregulatory range in Cerebral Circulation a wide



- Normally, blood flow to brain remains constant at perfusion pressures, **50-150 mm Hg**.
- In conditions of acute or chronic high blood pressures*, the autoregulatory curve range shifts to the right (as shown in the figure), thus, maintaining a normal blood flow even at higher pressures.
- Both the **myogenic and metabolic** mechanisms are involved in autoregulation.
- Both the **cerebral arteries and cerebral arterioles** can change their resistance proportionately during autoregulation.

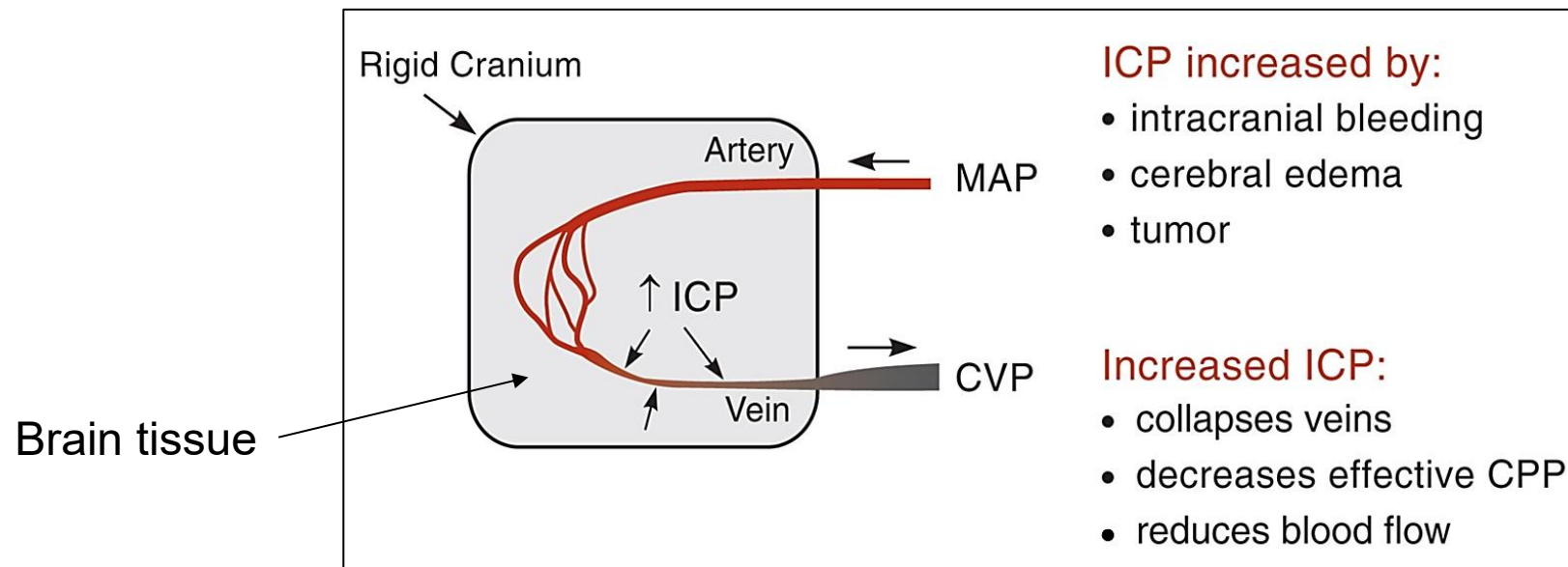
*Acute high pressure: in conditions of sympathetic stimulation;
Chronic high pressure: patients with long-standing hypertension.

Vasodilators of the Cerebral Circulation

- Major vasodilators of cerebral circulation are CO_2 and H^+ (source of H^+ is CO_2).
- $\uparrow \text{PCO}_2$ (hypercapnia) produces vasodilation and $\downarrow \text{PCO}_2$ (hypocapnia) causes vasoconstriction. (Ex: Voluntary hyperventilation may lead to light-headedness). Cerebral vessels are highly sensitive to a small change in PCO_2 .
- Vasodilator response to hypoxia (low PO_2) is strong only when PO_2 is below 50 mmHg.
- Other vasodilators include Potassium, adenosine and nitric oxide.
- Cerebral vessels are insensitive to circulating hormones because of the blood-brain barrier.
- Cerebral vessels do not respond to sympathetic nerve activity as they have sparse sympathetic innervations and lack alpha- or beta-adrenergic receptors.

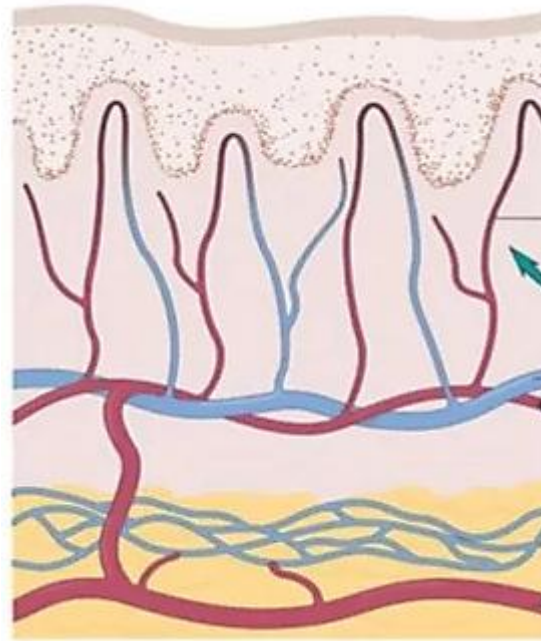
Effect of Intracranial pressure (ICP) on Cerebral Blood Flow

- ICP is the pressure within the rigid cranium (gray area in figure below).
- **Rise in ICP compresses** and collapses the low-pressure cerebral veins (arrows inside the gray area in the figure).
- **Cerebral perfusion pressure** (CPP) is the difference between mean arterial pressure (MAP) and the intracranial pressure. ($CPP = MAP - ICP$)*.
- Clinical conditions (listed in the figure below) that increase ICP, decrease the effective CPP and thus, also reduce cerebral blood flow.



*Note: in other organs, perfusion pressure is the difference between MAP and central venous pressure (CVP).

Cutaneous Circulation

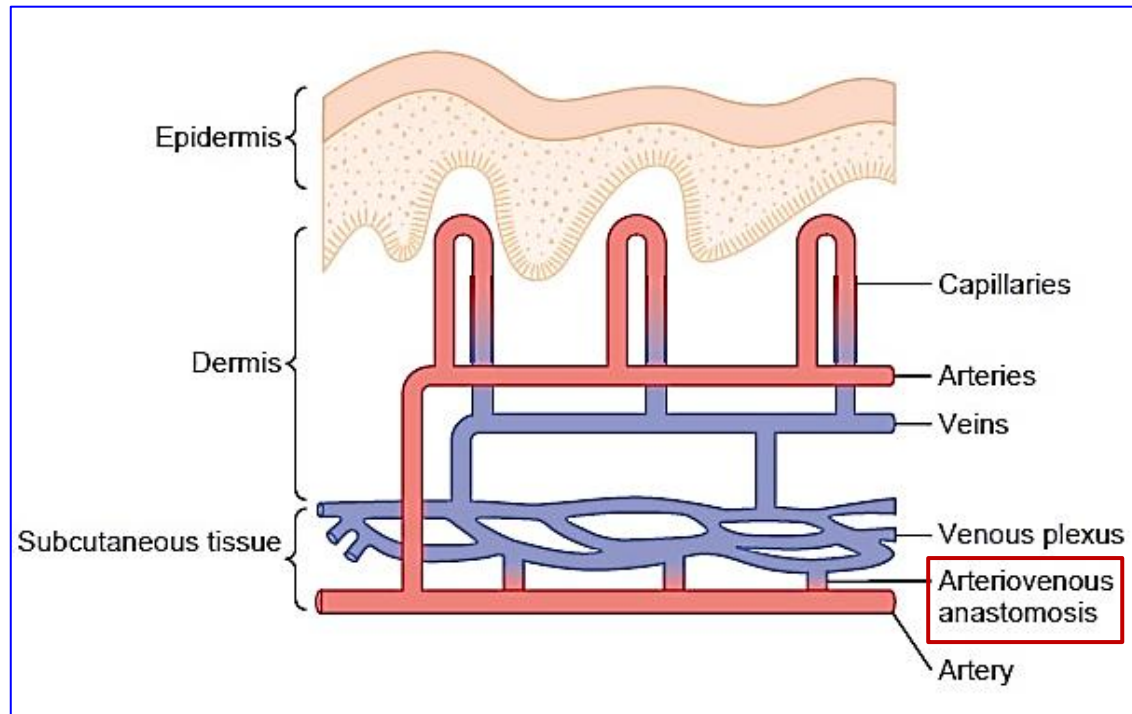


Cutaneous Circulation: Features

- Skin has a relatively low metabolic rate and receives about 4% of the cardiac output at rest.
- Skin blood flow can vary from 1% to 60% of the cardiac output under extreme cold and warm conditions, respectively.
- The wide variations in skin blood flow is the physiologic mechanism of heat transfer which is important for temperature regulation.
- Sympathetic nerves control skin blood flow (most important).
- Local regulatory mechanisms on blood flow and autoregulation are weak in skin.

Adjustment of cutaneous blood flow is used for body temperature regulation: Role of Arteriovenous anastomoses (AVA)

- Direct vascular connections between arteries and veins exist in the skin. These are known as **Arteriovenous anastomoses (AVA)**. These are richly innervated by sympathetic nerves.
- AVA are densely present in skin over the **hands, feet, ears, nose, and face**. They are less in skin over the limbs and torso.
- These AVA function as shunts: When AVA open, allow blood to flow directly into a large surface area of small venous vessels (venous plexus shown in the figure) just underlying the surface of the skin. Blood flow to skin decreases when these AVA are closed.



Both **sympathetic adrenergic** and **sympathetic cholinergic** nerves innervate arteriovenous anastomoses.

Adrenergic n activity: closes AVA;
Cholinergic n activity opens AVA.

Note: Adrenergic nerve releases Norepinephrine and cholinergic nerve releases Acetylcholine as neurotransmitters.

Adjustment of cutaneous blood flow during temperature regulation: Role of Arteriovenous anastomoses (AVA)

Exposure to **cold**



Activation of sympathetic **adrenergic** nerves



Closure of the arteriovenous anastomoses



Decreased blood flow to skin



Conservation of heat and minimize heat loss

Exposure to **heat**



Activation of sympathetic **cholinergic** nerves



Opening of the arteriovenous anastomoses



Increased blood flow to skin



Maximize heat loss and reduce heat conservation

Thermoregulatory centers located in the **Hypothalamus** control these selective activation of the sympathetic nerve fibers

Summary of Controls on the Special Circulations

	Autonomic control	Local control	Mechanical factors contributing control
Coronary	Not important	Very important as O ₂ delivery is flow-limited, Autoregulation via myogenic response, Metabolic hyperemia occurs.	Max blood flow during diastole. Coronary vessels are compressed during systole and thus, low blood flow.
Cerebral	Least important	Most important as neurons do not withstand, Autoregulation via myogenic response, Metabolic hyperemia occurs.	Flow is determined by intracranial pressure. Rise in intracranial pressure reduces blood flow to brain as in cerebral edema, brain tumor and head injury.
Cutaneous	Most Important: via sympathetic adrenergic and cholinergics	Weak autoregulation Weak reactive hyperemia	Arteriovenous anastomoses play important role in temperature regulation of the body.

Practice Question: Multiple choice type

A 24-year-old woman comes to the physician for a follow up examination. She has a 1-year history of panic disorder. Currently, she is not on any medication, but practices cognitive behavioral therapy during a panic attack. She says that she develops panic attacks when she thinks of her close friend's death in an accident. During such attacks, she develops palpitations, nausea, intense fear of dying, lightheadedness, chest pain, chills, choking, sweating, and she hyperventilates with an increased respiratory rate and depth. Which of the following mechanisms best explains lightheadedness during an episode in this patient?

- A. Cerebral vasoconstriction
- B. Cerebral vasodilation
- C. Increase in systemic blood pressure
- D. Release of catecholamines in blood
- E. Sympathetic nerve stimulation

Correct Answer: A. Hyperventilation causes hypocapnia ($\downarrow$ PCO₂) and cerebral vasoconstriction. CO₂ is a potential vasodilator in the cerebral circulation.

Practice Question: Multiple choice type

A group of volunteers participated in a study which exposed them to a period of cognitive stimulation such as reading, problem solving and talking. These activities result in increased neuronal activities and a significant increase in cerebral blood flow in focal brain areas. Which of the following sets of changes in cerebral tissue concentrations is the most likely explanation for the increase in cerebral blood flow?

	Carbon dioxide	pH	Potassium
A	↑	↑	↑
B	↑	↓	↑
C	↑	↓	↓
D	↑	↑	↓
E	↓	↓	↓
F	↓	↑	↓
G	↓	↑	↑
H	↓	↓	↑

Correct answer: B. Increased blood flow is because of the release of vasodilator substances like CO₂, H⁺ and potassium.

Practice Questions: Short answer type

- What would be the effect of a 10% drop in the partial pressure of O_2 in arterial blood on the cerebral circulation?
- What would be the effect of a 10% rise in the partial pressure of CO_2 in arterial blood on the cerebral circulation?
- Which of these two gases is most important for the regulation of cerebral blood flow under normal circumstances?

Answers: Practice Questions

- What would be the effect of a 10% drop in the partial pressure of O_2 in arterial blood on the cerebral circulation?
- Answer: Virtually no effect on cerebral blood flow. A significantly low PO_2 (below 50 mm Hg) is a strong vasodilator.
- What would be the effect of a 10% rise in the partial pressure of CO_2 in arterial blood on the cerebral circulation?
- Answer: A very significant increase in cerebral blood flow as the cerebral blood vessels are highly sensitive to a small change in PCO_2 .
- Which of these two gases is most important for the regulation of cerebral blood flow under normal circumstances?
- Answer: CO_2 . In some chronically hypercapnic patients the sensitivity to O_2 levels is increased and it becomes more dominant in the control of breathing and cerebral blood flow, but under normal circumstances CO_2 levels are the key control factor.

Any Questions ?

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